VCS VERIFICATION REPORT

SHANDONG PENGLAI PINGDINGSHAN WIND FARM PROJECT

(VCS PROJECT ID: 258)

Document Prepared By

Shenzhen CTI International Certification Co., Ltd

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Summary:

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions reported for the project activity “Shandong Penglai Pingdingshan Wind Farm Project” (VCS Project ID:258) for the monitoring period 01/01/2017-19/12/2018 within the 1st VCS crediting period 20/12/2008-19/12/2018, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.3.

The verification was performed on the basis of VCS Standard Version 4.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified one CL, and no CAR or FAR is raised in this monitoring period.

In CTI’s opinion, the GHG emission reductions reported for the project in the monitoring report (version 02 dated 08/09/2022) are fairly stated. The GHG emission reductions of the period from 01/01/2017-19/12/2018 within the 1st VCS crediting period were calculated correctly on the basis of approved methodology ACM0002 “Consolidated methodology for grid-connected electricity

generation from renewable sources" (version 07), the registered VCS PD i.e. CDM-PDD (version 05, dated 20/07/2009) and the revised monitoring plan approved by CDM EB on 24/02/2011.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

Hence, CTI is able to certify that the emission reductions from the "Shandong Penglai Pingdingshan Wind Farm Project" during this monitoring period from 01/01/2017-19/12/2018 amount to 239,704 tCO_2e.

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1 INTRODUCTION

Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd. has commissioned Shenzhen CTI International Certification Co., Ltd (CTI) to carry out the verification and certification of emission reductions reported for the "Shandong Penglai Pingdingshan Wind Farm Project" (the Project) for the monitoring period 01/01/2017-19/12/2018. This report contains the findings from the verification and includes a verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the "Shandong Penglai Pingdingshan Wind Farm Project" for the period 01/01/2017-19/12/2018.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan;
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide (version 4.2)/35/
- VCS Standard (version 4.3)/34/ and other relevant requirements defined by Verra;
- The approved methodology ACM0002 (version 07)/38/ applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.3, the project size categorization belongs to ‘Project’ as the expected annual emission reductions is 107,488 tCO_2e lower than 300,000 tCO_2e. Therefore, according to the requirement of para 4.1.8 (4) of VCS standard Version 4.3 “The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.”, CTI applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

1.4 Summary Description of the Project

Sectoral Scope and Project Type

According to the VCS Program Guide (version 4.2)/35/ and Annex A of the Kyoto Protocol, the project is applicable under the following activity categories:

- Sectoral Scope: 1. Energy (Renewable/non-renewable)
- Project Type: Grid-connected wind power project

Project Background

The project is located in Beigou Town, Penglai City, Shandong Province of P.R.China. The site location’s approximate coordinates are east longitude of 120°38'48"~120°42'11" and north latitude of 37°44'09"~37°46'17". The project activity was registered as a CDM project with Ref.2397 on 21/07/2009 and the renewable crediting period was selected starting on 21/07/2009. The project has been registered as VCS project with VCS ID. 258 under VCS Standard Version 2007 with the crediting period start date 20/12/2008 when the commencement of operations activities started as confirmed by checking the Commissioning Report /28/, operation logs/7/, the VCS monitoring report and verification report for the period of 20/12/2008-20/07/2009 /23//24/.
The amount of 71,328 tCO_2e VCUs have been issued for the monitoring period 20/12/2008-20/07/2009 within the 1st VCS crediting period. GHG emission reductions have been applied for CERs issuance under CDM of which total 245,407 tCO_2e have been issued for the monitoring periods: 21/07/2009-24/11/2009, 25/11/2009-24/02/2011 and 25/02/2011-25/10/2011.
The installed capacity of the project is 48MW equipped with 15 sets of wind turbine generator in type of V90-2.0MW, with unit capacity of 2MW and 10 set of wind turbine generator in type of V90-1.8MW, with unit capacity of 1.8MW, all manufactured by Vestas Wind Systems A/S The electricity generated by the project activity is supplied to the North China Power Grid (NCPG), and the project is estimated to deliver 107,488 tCO_2e emission reductions annually.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered VCS PD (version 05 dated 20/07/2009) and the revised monitoring plan approved by CDM EB on 24/02/2011, the applied methodology, and the VCS Standard (Version 4.3) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CTI appointed a verification team in accordance with CTI's internal procedures.

Function	Name	Technical competence	Task Performance*
Team Leader& Verifier	Feng Tian	1.2	☒DR ☒SV ☒RP ☐TR
Technical Reviewer	Lin Shunrong	1.2, 14.1, 15.1	☒DR ☐SV ☐RP ☒TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

In addition to the VER/VCU monitoring report/1/, the registered VCS PD /11/, emission reduction calculation spreadsheet/2/ during this monitoring period and revised monitoring plan approved by CDM EB on 24/02/2011, the following documents also were assessed as a part of the verification audit:

- The CDM registered PDD and corresponding validation report /11/;
- The previous VER/VCU monitoring report/23/ and verification report/24/
- The previous CDM/CER monitoring reports/25/ and verification reports/26/;
- Baseline and monitoring methodology ACM0002 (version 07) applied by the project/38/;

- Relevant decisions, clarifications and guidance from the Verra/34/-/37/; and
- Other information and references relevant to the project activity.

During the desk review, CTI has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and
- An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On 08/08/2022, CTI visited the project proponent China Resources Wind Power (Yantai) Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed or assisted the verification team /39/. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

The key personnel interviewed /39/ are summarized in the table below:

<table>
<tr><th>Interviewed personnel</th><th>Role</th><th>Organization</th><th>Subject</th></tr>
<tr><td>Mr. Wang Peide</td><td>Plant Director</td><td rowspan="3">China Resources Wind Power (Yantai) Co., Ltd.</td><td rowspan="3">Operation of the project activity;
Implementation of the monitor plan of the project activity;
Data collection and data achievement;
Calibration of meters and equipment maintenance;
The complaint by local stakeholders;
The stakeholder consultation during the operation of the project activity.
Project's sustainable development contributions.</td></tr>
<tr><td>Mr. Zhang Guanjun</td><td>O&M Staff</td></tr>
<tr><td>Mr. Qiu Jian</td><td>O&M Staff</td></tr>
<tr><td>Mr. Wei Shaoyuan</td><td rowspan="3">Local resident</td><td rowspan="3">Local Village</td><td rowspan="4">The impact of the project activity;
The complaint by local stakeholders;
The stakeholder consultation during the operation of the project activity;
Project's sustainable development contributions.</td></tr>
<tr><td>Mr. Gao Guanglong</td></tr>
<tr><td>Mr. Wang Mingdi</td></tr>
<tr><td>Mr. Lin Pengyu</td><td>Staff</td><td>Local Environment Protection Bureau</td></tr>
<tr><td>Ms. Ding Xue</td><td>Project Manager</td><td>Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.</td><td>Data collection and ER calculation.</td></tr>
</table>

2.4 Site Inspections

During the on-site assessment, CTI has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the registered VCS PD of the project activity;
- A review of information flows for generating, recording, aggregating and reporting the monitoring parameters; and
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the approved revised monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity receipts;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the approved revised monitoring plan and the selected methodology;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 07), and the management system were assessed during the site visit.

2.5 Resolution of Findings

A corrective action request (CAR)shall be raised, where:

i. Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;

ii. Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;

iii.Issues identified in a FAR during validation to be verified during verification have not been resolved by the project proponents.

A clarification request (CL) shall be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

The verification team identified 1 CL and no CAR in this monitoring period. The CL was satisfactorily addressed by the project proponents in the updated monitoring report version 02 dated 08/09/2022 (refer to Appendix D for further details).

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CTI confirmed that there was no FAR identified in previous verification/24/, and no FAR was raised during this verification.

2.6 Eligibility for Validation Activities

The project has been validated by TÜV NORD CERT GmbH under the CDM rules and registered as CDM project on 21/07/2009, and was registered as VCS project under VCS standard version 2007 through gap validation.

CTI was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the project. By checking the UNFCCC website, the VCS pipeline and the previous verification report, it is confirmed the verification for this monitoring period 01/01/2017-19/12/2018 which is no more than consecutive 6 years is the first verification undertaken by CTI. Therefore, the verification undertaken by CTI is in line with the requirement for VVB stipulated in Section 4.1.20 of VCS standard Version 4.3

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project has been registered as a CDM project on 21/07/2009 under the UNFCCC with Ref. 2397 The renewable CDM crediting period was selected staring on 21/07/2009. By checking the UNFCCC website: https://cdm.unfccc.int/Projects/DB/RWTUV1234176728.19/view, it is confirmed that total 245,407tCO_2e of CERs have been issued for the monitoring periods 21/07/2009-24/11/2009, 25/11/2009-24/02/2011 and 25/02/2011-25/10/2011.

3.2 Methodology Deviations

The validation process /16/ has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (version 07) /38/. There was not any methodology deviation applied to this project. Details refer to section 4.1.

3.3 Project Description Deviations

By checking the previous VCS verification reports with the monitoring period 20/12/2008-20/07/2009, the Commissioning report of the project /28/, operation logs/7/ and through the site visit interview with project owner, it is confirmed that the project started operation on 20/12/2008 which is the project start date according to the requirement of VCS. A deviation was identified for the crediting period of the Project. The project is registered as VCS project under VCS version 2007 and completed validation before 19/03/2020. As per VCS requirement, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice. Therefore, the first renewable crediting period of the Project under VCS is updated from 20/12/2008-20/07/2009 to 20/12/2008-19/12/2018, which lasts for 10 years. As per VCS requirement, the project did not conduct the validation of crediting period renewal within the two years after the end of the first VCS crediting period. Thus, the crediting period of the project ends on 19/12/2018.

CTI confirms that the project remains in compliance with VCS Program rules and the deviation on the crediting period has no impact on the applicability of the methodology, additionality of project activity, and the appropriateness of the baseline scenario.

By checking the previous verification reports and the UNFCCC website: https://cdm.unfccc.int/Projects/DB/RWTUV1234176728.19/view, a revised monitoring plan has been approved by CDM EB on 24/02/2011 due to that the Project shares the same 220kV substation (Tangqiu 220kV Substation) with the other wind farm, namely Shandong Penglai Wind Farm Phase II Project (Project B) which was put into operation since 12/11/2009. The revised monitoring plan specifies the monitoring and calculations of net electricity supplied to the grid by the project activity (EG_y) as follows:

EG_y is calculated by $EG_{export,total}$ - $EG_{import,total}$ - $EG_{export,other}$ and cross-check with $EG_{export,project}$ - $EG_{import,project}$., the conservative values will be applied for emission reductions calculation,

Where:

$EG_{export,total}$ is the electricity supplied by the Project and Project B to the grid;

$EG_{import,total}$ is the electricity imported from the grid to the Project and Project B;

$EG_{export,other}$ is the electricity supplied by the Project B to the grid;

$EG_{export,project}$ is electricity supplied by the project activity to the grid;

$EG_{import,project}$ is the electricity imported from the grid by the project.

The verification team confirmed that the deviations did not lower the accuracy and conservativeness of the monitoring system and will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals. Furthermore, the deviation relates only to the criteria and procedures for monitoring or measurement, doesn't relate to any other part of the methodology.

In conclusion, the verification team assessed through visual inspection and document review that all physical features of the project activity including wind turbines, generators, transmission facilities, monitoring equipment, data monitoring, reporting and collecting systems have been implemented in accordance with the registered PD and approved revised monitoring plan. CTI confirms that the project is implemented in accordance with the project description, and no any other project description deviations occurred. Details refer to section 4.1.

3.4 Grouped Project

The project was not a grouped project; hence this clause is not applicable.

4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the “Shandong Penglai Pingdingshan Wind Farm Project” for the monitoring period 01/01/2017-19/12/2018.

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

The project is a wind power grid-connected plant and locates at Beigou Town, Penglai City, Shandong Province of P.R.China. The geographical coordinates are verified by GPS system as east longitude of 120°38'48''~120°42'11'' and north latitude of 37°44'09''~37°46'17''. By checking the FSR /29/, equipment purchase contract/31/ and through on-site inspection, it is confirmed that the total installed capacity is 48MW equipped with 15 sets of wind turbine generator in type of V90-2.0MW, with unit capacity of 2MW and 10 set of wind turbine generator in type of V90-1.8MW, with unit capacity of 1.8MW. It was confirmed by checking the Commissioning report /28/, the operation logs/7/, the previous VCS monitoring report and verification report /23/-/24/ that the project was put into operation since 20/12/2008. The actual implementation of the project during this verification period was verified in terms of nameplates of wind turbines /27/ and the equipment purchase contract /31/. The details of the turbines with respect to their installation and capacity have been verified to be consistent with description indicated in the registered VCS PD.
The electricity generated by the project activity is supplied to the North China Power Grid, which can be confirmed by the Power purchase agreement (PPA) signed between China Resources Wind Power (Yantai) Co., Ltd. and Shandong Electric Power Company Yantai Branch Company/3/. All the monitoring system in operation period is consistent with the description in the monitoring plan. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the daily operation and maintenance records/7/ and via on-site visit, CTI can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including, recording and reporting was verified to be in place/6/ and their implementation was confirmed by interview with the key operators/39/ and observing the operation.

CTI confirms that the project implementation is in accordance with the project description contained in registered VCS PD. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the monitoring plan.

Participation or any rejection under any other GHG programs, any other form of environmental credit or emissions trading program since validation or previous verification

By checking the UNFCCC website: https://cdm.unfccc.int/Projects/DB/RWTUV1234176728.19/view, it is confirmed that the emission reductions within the monitoring periods 21/07/2009-24/11/2009, 25/11/2009-24/02/2011 and 25/02/2011-25/10/2011 have been issued as CERs. In addition, CDM-MR during the 4th monitoring period from 26/10/2011 to 24/06/2012 have been submitted to EB for publication but the request for issuance has not been submitted and the verification contract between PP and DOE (B.V) has been terminated as confirmed by checking the UNFCCC website. Via interviewing with PP and checking the Statement/32/ issued by PP, it is confirmed that the project proponent would not apply for the issuance of CERs for the period 26/10/2011 to 24/06/2012 and the monitoring report withdrawal request form has been submitted to EB by PP. The emission reductions during 26/10/2011 to 31/12/2016 is under VCS verification by another VVB as confirmed by interviewing with PP during on-site visit.

Therefore, it is concluded no double counting issuance has occurred for this verified monitoring period 01/01/2017-19/12/2018 under both VCS and CDM programmes. Via the on-site interview, the project proponent confirmed the emission reductions for this verified period are only requested for VCUs issuance.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction (CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification.
- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification.

The project is completely voluntary CDM and VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and energy intensive industries that emit at least 26,000 tonnes of CO_2e/year as per the regulation/19/ including thermal power generation, petrochemical, chemical, building materials, iron and steel, etc. By checking the business license/14/of the project proponent, it is confirmed that the PP is qualified in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no emission cap enforced for the PP by further checking the compulsive company list under China ETS in public information/20/.

Moreover, it is confirmed by CTI the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation/19/. Also, CCER mechanism has been suspended by Chinese government since 2017/20/, the project activity will not and is not allowed to registered under CCER mechanism. Thus, the assessment team concluded that the project activity not involved on other emissions trading programs and other binding limits.

The Statement on no double counting made by PP /32/ submitted by PP was checked by the verification team and confirmed to be correct in terms of the regulations and database of CCER and China ETS.

SD contributions resulted by implementation of the project activity

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- *SDG 7, Indicator 7.2.1 Renewable energy share in the total final energy consumption*
  The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in the gird North China Power Grid. During this monitoring period, 227,250.896MWh of net electricity generated by wind energy was delivered to the North China Power Grid. It is verified by checking the daily and monthly reading records/8/ and via on-site inspection. Meanwhile, by checking the approved VCS monitoring report for the previous monitoring, it is confirmed 294,873.196MWh of cumulative renewal electricity has been supplied to the grid since the project operation. This contributes to increasing the share of clean energy consumption of China's National Plan on Implementation of the 2030 Agenda for Sustainable Development /33/.
- *SDG 8, Indicator 8.5.2 Unemployment rate by sex, age and persons with disabilities*
  It is verified by interviewing with project proponent, staff and local villagers and checking the Staff list provided by PP /21/, that the project activity provides 26 long-term job opportunities for local residents including 6 positions for female which contributes to decrease the unemployment rate of the local community. Hence, the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable. This also contributes to "Increase labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment" as stated in China's National Plan on Implementation of the 2030 Agenda for Sustainable Development /33/.
- *SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed*
  As verified by on-site inspection, and via checking the emission reduction calculation sheet/2/ and the monthly reading records/8/, 239,704 tCO_2e GHG emission reductions are generated by the project activity during this monitoring period. And by checking the approved VCS monitoring report, it is confirmed the cumulative emission reductions are 311,032tCO_2e since operation of the project activity. This contributes achieve one of China's stated sustainable development priorities "Actively

adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas" stated in China's National Plan on Implementation of the 2030 Agenda for Sustainable Development /33/.

In conclusion, Verification Team was able to confirm that the project implementation is in accordance with the project description contained in the registered VCS PD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved VCS PD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and residents, noise, ambient air, sewage, solid waste, light pollution and electromagnetic radiation etc. is insignificant, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent, staff and local villagers and checking the Staff list /21/, that the project has contributed to the local employment by providing job opportunities for both males and females. As stated by the plant staff during the on-site interviews, they have gained knowledge and skills in the regular training organized by the project proponent, so it can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them.

In conclusion, the implementation of the project activity does not cause any significant negative environmental or socio-economic impacts and is beneficial to the local socio-economic environment.

4.2.2 Local Stakeholder Consultation

The local stakeholder's survey was carried out on 17/02/2008 by the project proponent around the project site. Total 50 questionnaires were given out and all of them were returned successfully.

In addition, a stakeholder meeting was held on 11/03/2008. 19 representatives of local stakeholders, including governmental officials of local county and local residents attended the meeting. The project owner introduced the proposed project and the opinions expressed by the stakeholders were recorded and are available on request.

The stakeholder meeting and the questionnaire survey showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.

For continuous communication with local stakeholders, through on-site inspection and interview during on-site visit, it is confirmed that the project owner has published its phone number of the central control room at the power plant gate to local people and village committees, anyone who has concerns can call the phone number to express their opinions directly. Also, the project proponent put a grievance book in the office of the wind farm and the villagers can leave their comments at any time. Through interviewing with staff from local environmental protection bureau and local residents during the site visit, it is confirmed by the verification team that during the implementation stage of this monitoring period, local authority has conducted spot checks on the implementation of the project periodically as per the request from the local governments' regulations, no negative comments nor grievance were received from the local stakeholders during this monitoring period were identified and the project passed all the periodic spot checks by local government.

All such conclusion has been verified through site visit and checking VCS PD.

4.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

Compliance of monitoring plan with monitoring methodology

CTI is able to confirm that the approved revised monitoring plan is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 07).

Compliance of monitoring with the monitoring plan

CTI confirms that the parameters stated in the monitoring plan are monitored and reported appropriately. Parameters required to be monitored by the approved revised monitoring plan as per the monitoring methodology ACM0002 (version 07) and the management system was assessed during the site visit. The monitoring report lists each parameter required by the approved revised monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for each parameter is provided. The information flow for each parameter in further verified in the following sections.

Parameters fixed ex ante

"Data and parameters fixed ex-ante" in the MR are checked against the registered VCS PD. The parameter $EF_{grid,CM,y}$ is the combined emission factor of the grid which was determined ex-ante at the validation stage and has been fixed during the first crediting period. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the registered VCS PD.

Other ex-ante parameters (i.e. $FC_{i,m,y}$, $NCV_{i,y}$, $EF_{CO2,i,y}$, Carbon Oxidation Factor, Installed Capacity, $GEN_{import,y}$, GEN_y and Efficiency level of the best technology commercially available for coal, oil and gas-fired power plant) were used to calculate the parameter $EF_{grid,CM,y}$ in the registered VCS PD, which were validated in the validation process. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the registered VCS PD.

Parameters monitored

CTI conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks, sales receipts and electricity confirmation;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the approved revised monitoring plan, the parameters to be monitored are as below:

$EG_{export,project}$	Electricity supplied by the project activity to the grid
$EG_{import,project}$	Electricity imported from the grid by the project
$EG_{export,total}$	Electricity supplied by the Project and Project B to the grid
$EG_{import,total}$	Electricity imported from the grid to the Project and Project B
$EG_{export,other}$	Electricity supplied by the Project B to the grid
EG_y	Net electricity supplied to the grid by the project activity

Monitored parameter $EG_{export,project}$, $EG_{import,project}$, $EG_{export,total}$, $EG_{import,total}$, $EG_{export,other}$ and EG_y

By site inspection and checking the registered monitoring plan and previous verification reports, CTI confirmed that the meters involved in the project are :

Main meter M1 (with accuracy class of 0.2S) and its backup meter M2 (with accuracy class of 0.5S) installed at Tangqiu Substation are owned by grid company to measure $EG_{export,total}$ and $EG_{import,total}$ and after confirmed by both sides, the sales receipts are issued based on the readings of M1 by the grid company.

Meters M4, M5, and M6 (all with accuracy class of 0.2S) installed on the collection line 1, 2, and 3 of the project are owned by PP to measure $EG_{export,project}$ and $EG_{import,project}$.

Meters M7, M8 and their backup meters M9, M10 (all with accuracy class of 0.5S) on the collection line 4 and 5 of Project B (Project B is owned by the same Project Proponent as this project by checking the Business license and interviewing with PP during on-site visit) are owned by PP to measure $EG_{export,other}$. After confirmed by both sides, sales receipts are issued by the grid company based on the readings of M7 and M8.

According to the approved revised monitoring plan, net electricity supplied to the grid by the project activity (EG_y) is calculated as follows:

$EG_y = EG_{export,total} - EG_{import,total} - EG_{export,other}$ and is cross checked with $EG_{export,project} - EG_{import,project}$.

The verification team had on-site checked the location of the meters against the diagram of power connection system /4/ and found them to be consistent and installed in accordance with the monitoring plan. All the monitoring facilities and system have been verified by CTI during on-site visit.

By checking operation log /7/ and via on-site visit, it is confirmed that all meters were working well during this monitoring period. Via the on-site interview and by checking the daily operation and maintenance records/7/, daily and monthly reading records/8/ electricity sales receipts /9/ and Electricity confirmation/10/ issued by grid company, it is confirmed by the verification team that $EG_{export,total}$ and $EG_{import,total}$ are continuously measured daily recorded and monthly aggregated by the gird company and PP at 24:00 on the last day of each month, after confirmed by both sides, the grid company supplies the reading records and the sales receipts of $EG_{export,total}$ and $EG_{import,total}$ to the project proponent every month and the sales receipts have been used for cross-check purpose. $EG_{export,other}$ is continuously measured, daily recorded and monthly aggregated by PP and grid company at 24:00 on the last day of each month. After confirmed by both sides, the sales receipts of $EG_{export,other}$ were issued and are used for cross-check purpose. $EG_{export,project}$ and $EG_{import,project}$ are continuously measured, daily recorded and monthly aggregated by PP at 24:00 on the last day of each month and the calculated differences between $EG_{export,project}$ and $EG_{import,project}$ are used to cross-check with EG_y calculated by $EG_{export,total} - EG_{import,total} - EG_{export,other}$.

Data in the monthly reading records were used to the report, through a cross check with sales receipts and electricity confirmation, the most conservative values were used for the monitoring report and ERs calculation spreadsheet, which has been verified by the verification team. Supporting references and data required to determine the quantity of net electricity generation supplied by the project is found to be complete and transparent.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the methodology ACM0002(version 07) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the approved revised monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

<table>
<tr><th>Monitoring equipment</th><th>Type</th><th>Serial NO.</th><th>Accuracy</th><th>Calibration frequency</th><th>Calibration date</th><th>Validity date</th><th>Calibration entity</th></tr>
<tr><td>M1 (old)</td><td>ZMQ202C</td><td>94089552</td><td>0.2S</td><td rowspan="10">At least annually</td><td>08/03/2016</td><td>07/03/2017</td><td rowspan="10">State Grid Shandong Electric Power Metering Centre</td></tr>
<tr><td>M1 (new)</td><td>ZMD402C</td><td>3730001000000206633042</td><td>0.2S</td><td>21/02/2017
07/03/2017
11/09/2017
06/03/2018
10/09/2018</td><td>20/02/2018
06/03/2018
10/09/2018
05/03/2019
09/09/2019</td></tr>
<tr><td>M2</td><td>DTSD341</td><td>20080778020034</td><td>0.5S</td><td>08/03/2016
07/03/2017
11/09/2017
06/03/2018
10/09/2018</td><td>07/03/2017
06/03/2018
10/09/2018
05/03/2019
09/09/2019</td></tr>
<tr><td>M4</td><td>DSTD341</td><td>09090167290077</td><td>0.2S</td><td rowspan="7">08/03/2016
07/03/2017
06/03/2018</td><td rowspan="7">07/03/2017
06/03/2018
05/03/2019</td></tr>
<tr><td>M5</td><td>DTSD341</td><td>09090167290078</td><td>0.2S</td></tr>
<tr><td>M6</td><td>DTSD341</td><td>09090167290079</td><td>0.2S</td></tr>
<tr><td>M7</td><td>DSTD341</td><td>200908326835</td><td>0.5S</td></tr>
<tr><td>M8</td><td>DSTD341</td><td>200908326832</td><td>0.5S</td></tr>
<tr><td>M9</td><td>DSTD341</td><td>200908326874</td><td>0.5S</td></tr>
<tr><td>M10</td><td>DSTD341</td><td>200908326873</td><td>0.5S</td></tr>
</table>

During this monitoring period, the old meter M1 (SN. 94089552) was replaced with the new M1 (SN. 3730001000000206633042) on 03/03/2017. CTI confirmed through on-site interviews with PP and by reviewing the qualification documents of State Grid Shandong Electric Power Metering Centre/12/and the meter calibration reports/13/ that the meter replacement was carried out with the consent by the grid company and the new meter's accuracy is 0.2S which is the same as the old meter and thus, still met the accuracy requirements of the monitoring plan.

Calibration records and accreditation certificates/12//13/have been verified by the verification team. The actual accuracy for M1(old and new), M4-M6 is verified to be 0.2S and the accuracy for

M2, M7-M10 is verified to be 0.5S which are in line with the requirement in the approved revised monitoring plan. The accuracy of all meters is in compliance with the Technical administrative code of electric energy metering /15/. The calibration frequency of all meters is at least annual, which is consistent with the monitoring plan. Based on the site visit and by checking the calibration reports, CTI confirms that:

- All meters have been installed in accordance with the approved revised monitoring plan;
- The meters' accuracy is in line with the requirement of the approved revised monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the approved revised monitoring plan and national technical standard/15/ as required by VCS standard. And the calibrations are verified to be valid for the whole reporting period.

CL01 was raised and successfully closed, please refer to Appendix D for details.

Data management and control

China Resources Wind Power (Yantai) Co., Ltd. is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the project management and monitoring manual/5/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/39/ and check records /7//8/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and approved revised monitoring plan and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

Emission reduction calculation

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$ER_y = BE_y - PE_y - L_y$

Baseline emissions

Baseline emissions are determined as multiplying quantity of net electricity supplied to the Grid by the project (EG_y) by the validated ex-ante fixed grid emission factor ($EF_{grid,CM,y}$).

$BE_y = EG_y \times EF_{grid,CM,y}$

Combined margin CO_2 emission factor ($EF_{grid,CM,y}$)

$EF_{grid,CM,y}$ is the grid emission factor of the which has been verified ex-ante in the validation stage in the VCS PD as 1.0548 tCO_2e/MWh.

Net electricity supplied to the Grid by the project ($EG_{facility,y}$)

As per the Section 4.1 above, EG_y is calculated as:
$EG_{export,total}$ - $EG_{import,total}$ - $EG_{export,other}$ and cross-check with $EG_{export,project}$ - $EG_{import,project}$.
The verification team has checked the daily and monthly reading records, sales receipts and electricity confirmation issued by the grid company and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The monthly electricity data and EG_y calculation are summarized in the Tables as follow:

Table 1 Monitored monthly values of $EG_{export,total}$ and $EG_{import,total}$

Monitoring period	$EG_{export,total}$			$EG_{import,total}$		
	Data based on meter readings M1	Data based on sales receipt	Conservative value	Data based on meter readings M1	Data based on sales receipt	Conservative value
	MWh	MWh	MWh	MWh	MWh	MWh
01/01/2017-31/01/2017	16,576.820	16,576.820	16,576.820	54.120	54.120	54.120
01/02/2017-28/02/2017	21,275.500	21,275.500	21,275.500	63.096	63.096	63.096
01/03/2017-31/03/2017	15,284.020	15,284.020	15,284.020	39.600	39.600	39.600
01/04/2017-30/04/2017	18,202.270	18,202.270	18,202.270	60.456	60.456	60.456
01/05/2017-31/05/2017	23,603.980	23,603.980	23,603.980	22.440	22.440	22.440
01/06/2017-30/06/2017	18,457.300	18,457.300	18,457.300	40.656	40.656	40.656
01/07/2017-31/07/2017	12,610.490	12,610.490	12,610.490	48.312	48.312	48.312
01/08/2017-31/08/2017	10,618.080	10,618.080	10,618.080	144.408	144.408	144.408
01/09/2017-30/09/2017	9,889.970	9,889.970	9,889.970	96.096	96.096	96.096
01/10/2017-31/10/2017	16,231.510	16,231.510	16,231.510	73.128	73.128	73.128
01/11/2017-30/11/2017	28,383.170	28,383.170	28,383.170	26.664	26.664	26.664
01/12/2017-31/12/2017	21,565.290	21,565.290	21,565.290	34.452	34.452	34.452
Subtotal 2017	-	-	**212,698.400**	-	-	**703.428**
01/01/2018-31/01/2018	22,763.480	22,763.480	22,763.480	28.987	28.987	28.987
01/02/2018-28/02/2018	20,463.010	20,463.010	20,463.010	50.952	50.952	50.952
01/03/2018-31/03/2018	31,106.960	31,106.960	31,106.960	27.799	27.799	27.799
01/04/2018-30/04/2018	31,410.220	31,410.220	31,410.220	26.875	26.875	26.875
01/05/2018-31/05/2018	19,215.400	19,215.400	19,215.400	46.174	46.174	46.174
01/06/2018-30/06/2018	20,030.230	20,030.230	20,030.230	96.096	96.096	96.096
01/07/2018-31/07/2018	21,259.180	21,259.180	21,259.180	58.792	58.792	58.792
01/08/2018-31/08/2018	13,236.670	13,236.670	13,236.670	89.074	89.074	89.074
01/09/2018-30/09/2018	9,462.870	9,462.870	9,462.870	84.665	84.665	84.665
01/10/2018-31/10/2018	18,576.970	18,576.970	18,576.970	72.257	72.257	72.257
01/11/2018-30/11/2018	13,403.990	13,403.990	13,403.990	77.933	77.933	77.933
01/12/2018-19/12/2018	11,813.195	11,813.195	11,813.195	30.112	30.112	30.112
Subtotal 2018	-	-	**232,742.175**	-	-	**689.716**

Total	-	-	445,440.575	-	-	1,393.144

Note:

1. As the cut-off time is 24:00 on the last day of each month, monthly readings of $EG_{export,total}$ and $EG_{import,total}$ are cross-checked with the sales receipts issued by grid company, the most conservative values are applied for calculation of EG_{y}.

2. Readings of $EG_{export,total}$ and $EG_{import,total}$ on the last day of this monitoring period i.e. 19/12/2018 are sourced from the daily reading records and have been verified with the confirmation/10/ issued by the grid company

Table 2 Monitored monthly values of $EG_{export,other,y}$ and EG_{y} calculated by $EG_{export,total}$- $EG_{import,total}$- $EG_{export,other}$

Monitoring period	$EG_{export,other,y}$					EG_{y}
	Data based on meter readings			Data based on sales receipt	Conservative value	Calculated by $EG_{export,total}$- $EG_{import,total}$- $EG_{export,other}$
	Meter readings based on M7	Meter readings based on M8	Subtotal			
	MWh	MWh	MWh	MWh	MWh	MWh
01/01/2017-31/01/2017	4,604.600	3,382.400	7,987.000	7,987.000	7,987.000	8,535.700
01/02/2017-28/02/2017	5,827.850	4,654.440	10,482.290	10,482.290	10,482.290	10,730.114
01/03/2017-31/03/2017	4,302.550	3,405.080	7,707.630	7,707.630	7,707.630	7,536.790
01/04/2017-30/04/2017	5,062.050	3,885.000	8,947.050	8,947.050	8,947.050	9,194.764
01/05/2017-31/05/2017	6,653.500	4,767.280	11,420.780	11,420.780	11,420.780	12,160.760
01/06/2017-30/06/2017	5,026.700	3,811.920	8,838.620	8,838.620	8,838.620	9,578.024
01/07/2017-31/07/2017	3,588.200	2,503.480	6,091.680	6,091.680	6,091.680	6,470.498
01/08/2017-31/08/2017	3,106.950	1,971.760	5,078.710	5,078.710	5,078.710	5,394.962
01/09/2017-30/09/2017	2,772.700	1,986.320	4,759.020	4,759.020	4,759.020	5,034.854
01/10/2017-31/10/2017	4,452.000	3,275.720	7,727.720	7,727.720	7,727.720	8,430.662

01/11/2017-30/11/2017	7,919.450	5,952.240	13,871.690	13,871.690	13,871.690	14,484.816
01/12/2017-31/12/2017	6,136.550	4,135.880	10,272.430	10,272.430	10,272.430	11,258.408
Subtotal 2017	-	-	-	-	**103,184.620**	**108,810.352**
01/01/2018-31/01/2018	6,124.300	4,615.240	10,739.540	10,739.540	10,739.540	11,994.953
01/02/2018-28/02/2018	5,780.950	4,509.400	10,290.350	10,290.350	10,290.350	10,121.708
01/03/2018-31/03/2018	8,406.650	7,108.640	15,515.290	15,515.290	15,515.290	15,563.871
01/04/2018-30/04/2018	8,543.150	6,995.520	15,538.670	15,538.670	15,538.670	15,844.675
01/05/2018-31/05/2018	5,165.650	4,047.960	9,213.610	9,213.610	9,213.610	9,955.616
01/06/2018-30/06/2018	5,550.650	4,103.120	9,653.770	9,653.770	9,653.770	10,280.364
01/07/2018-31/07/2018	6,111.000	4,217.360	10,328.360	10,328.360	10,328.360	10,872.028
01/08/2018-31/08/2018	3,839.150	2,745.400	6,584.550	6,584.550	6,584.550	6,563.046
01/09/2018-30/09/2018	2,711.800	1,768.480	4,480.280	4,480.280	4,480.280	4,897.925
01/10/2018-31/10/2018	5,249.650	3,832.920	9,082.570	9,082.570	9,082.570	9,422.143
01/11/2018-30/11/2018	3,839.150	2,602.320	6,441.470	6,441.470	6,441.470	6,884.587
01/12/2018-19/12/2018	3,390.642	2,352.813	5,743.455	5,743.455	5,743.455	6,039.628
Subtotal 2018	-	-	-	-	**113,611.915**	**118,440.544**
Total	-	-	-	-	**216,796.535**	**227,250.896**

Note:

1. As the cut-off time is 24:00 on the last day of each month, monthly readings of $EG_{export,other}$ are cross-checked with the sales receipts issued by grid company, the most conservative values are applied for calculation of EG_y.

2. Reading of $EG_{export,other}$ on the last day of this monitoring period i.e. 19/12/2018 are sourced from the daily reading records and have been verified with the confirmation/10/ issued by the grid company.

Table 3 Monitored monthly values of $EG_{export,project}$, $EG_{import,project}$ and EG_y calculated by $EG_{export,project}$ - $EG_{import,project}$

Monitoring period	$EG_{export,project}$				$EG_{import,project}$				EG_y
	Data based on meter readings M4	Data based on meter readings M5	Data based on meter readings M6	Subtotal	Data based on meter readings M4	Data based on meter readings M5	Data based on meter readings M6	Subtotal	Calculated by $EG_{export,project}$- $EG_{import,project}$
	MWh	MWh	MWh	MWh	MWh	MWh	MWh	MWh	MWh
01/01/2017-31/01/2017	2,494.380	4,167.044	2,072.084	8,733.508	6.972	10.388	5.628	22.988	8,710.520
01/02/2017-28/02/2017	2,955.512	5,288.164	2,737.532	10,981.208	4.844	6.944	3.080	14.868	10,966.340
01/03/2017-31/03/2017	2,124.360	3,651.984	1,957.144	7,733.488	6.356	8.820	4.788	19.964	7,713.524
01/04/2017-30/04/2017	2,492.560	4,538.016	2,287.656	9,318.232	10.976	18.620	8.008	37.604	9,280.628
01/05/2017-31/05/2017	3,323.320	6,050.520	3,050.040	12,423.880	3.920	6.720	2.240	12.880	12,411.000
01/06/2017-30/06/2017	2,602.600	4,790.800	2,333.240	9,726.640	5.600	1.400	3.080	10.080	9,716.560
01/07/2017-31/07/2017	1,811.600	3,300.360	1,499.120	6,611.080	13.720	21.000	10.080	44.800	6,566.280
01/08/2017-31/08/2017	1,597.680	2,735.880	1,342.600	5,676.160	14.840	23.800	12.320	50.960	5,625.200
01/09/2017-30/09/2017	1,477.280	2,537.067	1,230.587	5,244.934	9.520	15.960	8.400	33.880	5,211.054
01/10/2017-31/10/2017	2,447.480	4,132.788	2,068.346	8,648.614	8.120	11.480	6.440	26.040	8,622.574
01/11/2017-30/11/2017	4,068.120	7,121.520	3,500.000	14,689.640	2.800	3.920	1.960	8.680	14,680.960
01/12/2017-31/12/2017	3,235.680	5,555.480	2,665.320	11,456.480	4.480	6.720	3.640	14.840	11,441.640
Subtotal 2017	-	-	-	**111,243.864**	-	-	-	**297.584**	**110,946.280**
01/01/2018-31/01/2018	3,351.040	5,851.160	2,986.480	12,188.680	5.040	7.000	3.640	15.680	12,173.000
01/02/2018-28/02/2018	2,840.320	4,802.840	2,675.120	10,318.280	4.760	6.720	3.640	15.120	10,303.160
01/03/2018-31/03/2018	4,361.840	7,417.200	4,082.120	15,861.160	4.760	6.440	3.360	14.560	15,846.600
01/04/2018-30/04/2018	4,271.680	7,733.600	4,075.400	16,080.680	3.920	5.040	2.800	11.760	16,068.920
01/05/2018-31/05/2018	2,753.520	4,909.800	2,482.480	10,145.800	6.720	11.760	6.440	24.920	10,120.880
01/06/2018-30/06/2018	2,772.000	5,321.960	2,527.280	10,621.240	12.040	19.320	10.360	41.720	10,579.520
01/07/2018-31/07/2018	2,796.360	5,820.920	2,504.040	11,121.320	10.360	14.280	6.720	31.360	11,089.960
01/08/2018-31/08/2018	1,920.800	3,377.080	1,539.160	6,837.040	9.240	12.880	6.440	28.560	6,808.480
01/09/2018-30/09/2018	1,436.960	2,521.400	1,166.480	5,124.840	8.400	14.560	7.560	30.520	5,094.320

01/10/2018-31/10/2018	2,609.040	4,740.960	2,331.840	9,681.840	8.120	12.600	7.280	28.000	9,653.840
01/11/2018-30/11/2018	2,051.560	3,399.480	1,674.960	7,126.000	7.840	12.040	6.720	26.600	7,099.400
01/12/2018-19/12/2018	1,794.556	2,915.188	1,480.333	6,190.077	2.574	3.261	2.059	7.894	6,182.183
Subtotal 2018	-	-	-	**121,296.957**	-	-	-	**276.724**	**121,020.263**
Total	-	-	-	**232,540.821**	-	-	-	**574.278**	**231,966.543**

Note:

1. As the cut-off time is 24:00 on the last day of each month, monthly readings of $EG_{export,project}$ and $EG_{import,project}$ are checked by CTI with the original copy of the reading records during on-site visit and confirmed that the values of $EG_{export,project}$ and $EG_{import,project}$ used to calculate ERs are correct and complete.

2. Reading of $EG_{export,project}$ and $EG_{import,project}$ on the last day of this monitoring period i.e. 19/12/2018 are sourced from the daily reading records and have also been verified with the original copy of reading records by the verification team during on-site visit.

Table 4 Cross-check of EG_y and the Conservative values of EG_y

Monitoring period	EG_y		
	Calculated by $EG_{export,\ total}$- $EG_{impot,total}$- $EG_{export,other,y}$	Calculated by $EG_{export,project}$- $EG_{import,project}$	Conservative values
	MWh	MWh	MWh
01/01/2017-31/01/2017	8,535.700	8,710.520	8,535.700
01/02/2017-28/02/2017	10,730.114	10,966.340	10,730.114
01/03/2017-31/03/2017	7,536.790	7,713.524	7,536.790
01/04/2017-30/04/2017	9,194.764	9,280.628	9,194.764
01/05/2017-31/05/2017	12,160.760	12,411.000	12,160.760
01/06/2017-30/06/2017	9,578.024	9,716.560	9,578.024
01/07/2017-31/07/2017	6,470.498	6,566.280	6,470.498
01/08/2017-31/08/2017	5,394.962	5,625.200	5,394.962
01/09/2017-30/09/2017	5,034.854	5,211.054	5,034.854
01/10/2017-31/10/2017	8,430.662	8,622.574	8,430.662
01/11/2017-30/11/2017	14,484.816	14,680.960	14,484.816
01/12/2017-31/12/2017	11,258.408	11,441.640	11,258.408

Subtotal 2017	**108,810.352**	**110,946.280**	**108,810.352**
01/01/2018-31/01/2018	11,994.953	12,173.000	11,994.953
01/02/2018-28/02/2018	10,121.708	10,303.160	10,121.708
01/03/2018-31/03/2018	15,563.871	15,846.600	15,563.871
01/04/2018-30/04/2018	15,844.675	16,068.920	15,844.675
01/05/2018-31/05/2018	9,955.616	10,120.880	9,955.616
01/06/2018-30/06/2018	10,280.364	10,579.520	10,280.364
01/07/2018-31/07/2018	10,872.028	11,089.960	10,872.028
01/08/2018-31/08/2018	6,563.046	6,808.480	6,563.046
01/09/2018-30/09/2018	4,897.925	5,094.320	4,897.925
01/10/2018-31/10/2018	9,422.143	9,653.840	9,422.143
01/11/2018-30/11/2018	6,884.587	7,099.400	6,884.587
01/12/2018-19/12/2018	6,039.628	6,182.183	6,039.628
Subtotal 2018	**118,440.544**	**121,020.263**	**118,440.544**
Total	**227,250.896**	**231,966.543**	**227,250.896**

The corresponding baseline emissions during this monitoring period is calculated as:

Year	EG_y (MWh)	$EF_{grid,CM,y}$ (tCO_2e/MWh)	BE_y (tCO_2e)
2017 (01/01/2017-31/12/2017)	108,810.352	1.0548	114,773
2018 (01/12/2018-19/12/2018)	118,450.544	1.0548	124,931
Total	**227,250.896**	-	**239,704**

Project emissions

As statement in the registered VCS PD, for the wind power activities, the project emissions from the project is zero. Hence, PE_y during the monitoring period from Shandong Penglai Pingdingshan Wind Farm Project is zero.

Leakages

Leakage does not need to be accounted for this project as per the monitoring methodology ACM0002 (version 07).

Emission reductions

The emission reductions for this monitoring period were calculated as:

Monitoring period	GHG emission reductions or removals (tCO_2e)
01/01/2017 – 31/12/2017	114,773
01/01/2018 – 19/12/2018	124,931
Total ERs claimed	**239,704**

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

The assessment of each parameter related to calculate the GHG emission reductions is justified in Sections 4.1 and 4.4 above. Via the on-site inspection and desk review, the verification team verified the supporting evidence provided by the project proponent as listed in Appendix 3 used to determine the GHG emission reductions

For each reported data, the evidence is provided and verified as sufficient and quality is appropriate. Also, the cross-checks have been performed on the reported data with different source of evidence. The information flow from data generation and aggregation, to recording, calculation and final transposition into the monitoring report has been assessed by VVB in Section 4.4. Moreover, the calibration is verified as stated in Section 4.1 via on-site inspection with evidences to confirm the meters' location, accuracy and calibration frequency are in line with requirement of the monitoring plan.

Therefore, it is concluded that the evidence provided are verified as sufficient and quality is appropriate and thus the evidence can be used to determine the GHG emission reductions and removals for this monitoring period.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions that have been reported for the project activity "Shandong Penglai Pingdingshan Wind Farm Project" in China (VCS Project ID: 258) for the period 01/01/2017-19/12/2018.

The verification is based on the baseline and monitoring methodology ACM0002 (version 07), the VCS PD. The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the approved revised monitoring plan and the reporting of GHG emission reductions on the basis set out within the project monitoring report.

Our verification and validation approach was based on the requirements as defined under the applicable VCS Standard Version 4.3 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered VCS PD ;
- the monitoring plan is as per the applied methodology;
- the monitoring complies with the monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard Version 4.3 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.
- the project deviation is appropriately described and justified; the deviation was verified by the verification team as valid, and the project remains in compliance with the VCS rules.

It is CTI's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "Shandong Penglai Pingdingshan Wind Farm Project" for the period 01/01/2017-19/12/2018 are fairly stated in the monitoring report (version 02 dated 08/09/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 07) and the monitoring plan contained in the registered VCS PD.

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Reporting period: 01/01/2017-19/12/2018

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2017 (01/01/2017-31/12/2017)	114,773	0	0	114,773
2018 (01/02/2018-19/12/2018)	124,931	0	0	124,931
Total	**239,704**	**0**	**0**	**239,704**

Feng Tian

Mr. Feng Tian

Team Leader

19-September-2022

Shunrong Lin

Ms. Lin Shunrong

Technical Reviewer

19-September-2022

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CCER	Chinese Certified Emission Reduction
CER	Certified Emission Reduction(s)
CL	Clarification request
CO_2	Carbon Dioxide
CO_2e	Carbon Dioxide Equivalent
CTI	Shenzhen CTI International Certification Co., Ltd
DOE	Designated Operational Entity
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETN	Electricity Transaction Note
EU ETS	EU Emissions Trading Scheme
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse Gas(es)
IREC	International Renewable Energy Certificates
MR	Monitoring Report
NCPG	North China Power Grid
PD	Project Description
PP	Project Proponent
SN	Serial Number
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCES

Documentation used to verify the information provided by the project proponents

/1/	Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.: VER/VCU Monitoring Report for Shandong Penglai Pingdingshan Wind Farm Project, version 01 dated 01/08/2022, version 02 dated 08/09/2022
/2/	Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.: Emission reduction calculation spreadsheet for Shandong Penglai Pingdingshan Wind Farm Project, version 02 dated 08/09/2022
/3/	China Resources Wind Power (Yantai) Co., Ltd. and Shandong Electric Power Company Yantai Branch Company: Power Purchase Agreement for Shandong Penglai Pingdingshan Wind Farm Project
/4/	China Resources Wind Power (Yantai) Co., Ltd.： Diagram of power connection system
/5/	China Resources Wind Power (Yantai) Co., Ltd.: VER monitoring manual and management procedure.
/6/	China Resources Wind Power (Yantai) Co., Ltd.: Records of training for on-site staff.
/7/	China Resources Wind Power (Yantai) Co., Ltd.: Daily operation and maintenance records.
/8/	China Resources Wind Power (Yantai) Co., Ltd.: Daily and monthly reading records of Shandong Penglai Pingdingshan Wind Farm Project from 01/01/2017-19/12/2018
/9/	Shandong Electric Power Company Yantai Branch Company.: Electricity Sales Receipts from 01/01/2017-19/12/2018
/10/	Shandong Electric Power Company Yantai Branch Company: Electricity confirmation
/11/	China Resources Wind Power (Yantai) Co., Ltd.: Registered VSC PD i.e. CDM-PDD (version 05 dated 20/07/2009) TÜV NORD CERT GmbH: CDM Validation Report, Rev. NO.0.3, dated 20/07/2009
/12/	China National Accreditation Service for Conformity Assessment: Accreditation certificates of State Grid Shandong Electric Power Metering Centre which validity covering this monitoring period 01/01/2017-19/12/2018
/13/	State Grid Shandong Electric Power Metering Centre: Calibration certificates for meters: M1 (old and new), M2, M4-M10 covering this monitoring period
/14/	Business license of China Resources Wind Power (Yantai) Co., Ltd.

/15/	State Economic and Trade Commission: Technical administrative code of electric energy metering (DL/T 448-2016)
/16/	UNFCCC, http://cdm.unfccc.int
/17/	Gold Standard, http://www.goldstandard.org/
/18/	China CCER Exchange Info-platform, http://cdm.ccchina.org.cn/ccer.aspx
/19/	Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020
/20/	Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China
/21/	China Resources Wind Power (Yantai) Co., Ltd.: Staff list of the project
/22/	The revised monitoring plan approved by CDM EB on 24/02/2011 Validation report of revised monitoring plan
/23/	VER/VCU Monitoring Report for Shandong Penglai Pingdingshan Wind Farm Project of previous monitoring period 20/12/2008-20/07/2009
/24/	VCS Verification Report for Shandong Penglai Pingdingshan Wind Farm Project of previous monitoring period 20/12/2008-20/07/2009
/25/	CDM/CER Monitoring Reports for Shandong Penglai Pingdingshan Wind Farm Project of previous CDM monitoring periods 21/07/2009-24/11/2009, 25/11/2009-24/02/2011 and 25/02/2011-25/10/2011
/26/	CDM Verification Reports for Shandong Penglai Pingdingshan Wind Farm Project of previous monitoring period 21/07/2009-24/11/2009, 25/11/2009-24/02/2011 and 25/02/2011-25/10/2011
/27/	Vestas Wind Systems A/S: Nameplates of the equipment
/28/	China Resources Wind Power (Yantai) Co., Ltd.: Commissioning report
/29/	Feasibility Study Report of Shandong Penglai Pingdingshan Wind Farm Project
/30/	EIA Report of Shandong Penglai Pingdingshan Wind Farm Project dated 08/2006 EIA approval issued on 23/10/2007
/31/	Co-signed by the Vestas Wind Systems A/S, and China Resources Wind Power (Yantai) Co., Ltd.: Wind Turbines Purchase Agreement
/32/	China Resources Wind Power (Yantai) Co., Ltd.: Statement on no double counting

/33/	China's National Plan on Implementation of the 2030 Agenda for Sustainable Development: http://www.gov.cn/xinwen/2016-10/13/5118514/files/44cb945589874551a85d49841b568f18.pdf

Methodologies, tools and other guidance

/34/	Verified Carbon Standard: VCS Standard, Version 4.3
/35/	Verified Carbon Standard: VCS Program Guide, version 4.2
/36/	Verified Carbon Standard: VCS Sectoral Scopes http://v-c-s.org/node/448
/37/	Verified Carbon Standard: Registration and Issuance Process, version 4.2
/38/	UNFCCC EB: Approved methodology, ACM0002 , version 07

Persons interviewed

	Interviewed personnel	Role	Organization
/39/	Mr. Wang Peide	Plant Director	China Resources Wind Power (Yantai) Co., Ltd.
	Mr. Zhang Guanjun	O&M Staff	
	Mr. Qiu Jian	O&M Staff	
	Mr. Wei Shaoyuan	Local resident	Local Village
	Mr. Gao Guanglong		
	Mr. Wang Mingdi		
	Mr. Lin Pengyu	Staff	Local Environment Protection Bureau
	Ms. Ding Xue	Project Manager	Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.

APPENDIX C: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Mr. Tian FENG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	-	√	√	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by: Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 10/02/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by: Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

APPENDIX D: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 01	MR (version 01 dated 01/08/2022) lacks the information on calibration date and validity of the electricity meters, thus CL01 was raised to clarify.	The information on calibration of the meters had been included in the updated in the MR.	OK. It has been revised and verified by CTI by checking updated MR and the calibration reports. Therefore, CL 01 is closed.

Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA